International and Monetary Economics

Course Code: 30054

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Lecture 4

Roadmap

- The Nominal Exchange Rate
  1. Definitions
  2. Foreign Exchange Markets
  3. The Demand for Foreign Currency Assets
  4. Equilibrium in the Foreign Exchange Market

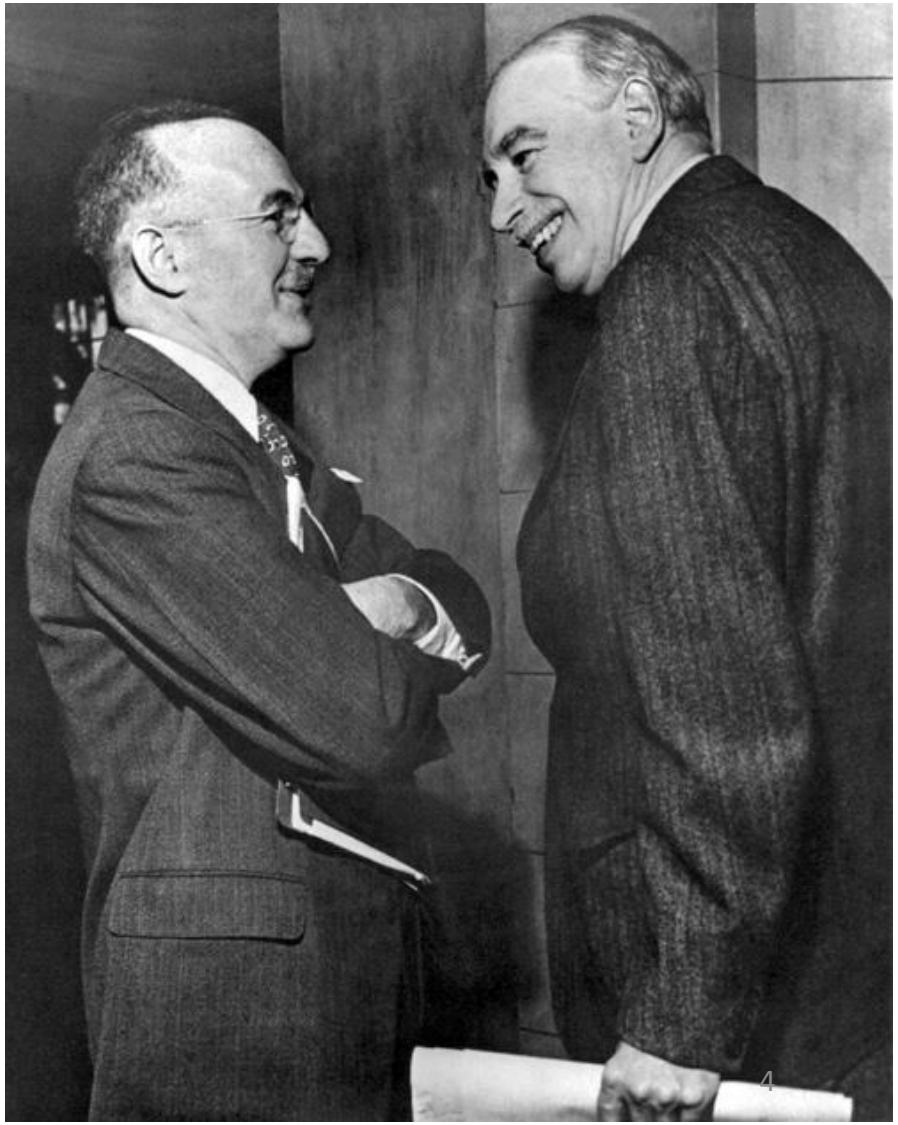
The Nominal Exchange Rate

John Maynard Keynes

1883 - 1946

Famous for

- Keynesian macroeconomics
- An architect of the Bretton Woods system (including the creation of the IMF)
- Made fortunes trading bonds, stocks, commodities
- Coined “interest rate parity” term (later today)
- Keynes was less successful with FX markets!



The Nominal Exchange Rate

Acquista iPhone 17 Pro

€ 2.489,00

Buy iPhone 17 Pro

$1,999.00

CCY - Delayed Quote • USD

EUR/USD (EURUSD=X)

1.1853 -0.0013 (-0.11%)

As of 4:29:43 PM GMT. Market Open.

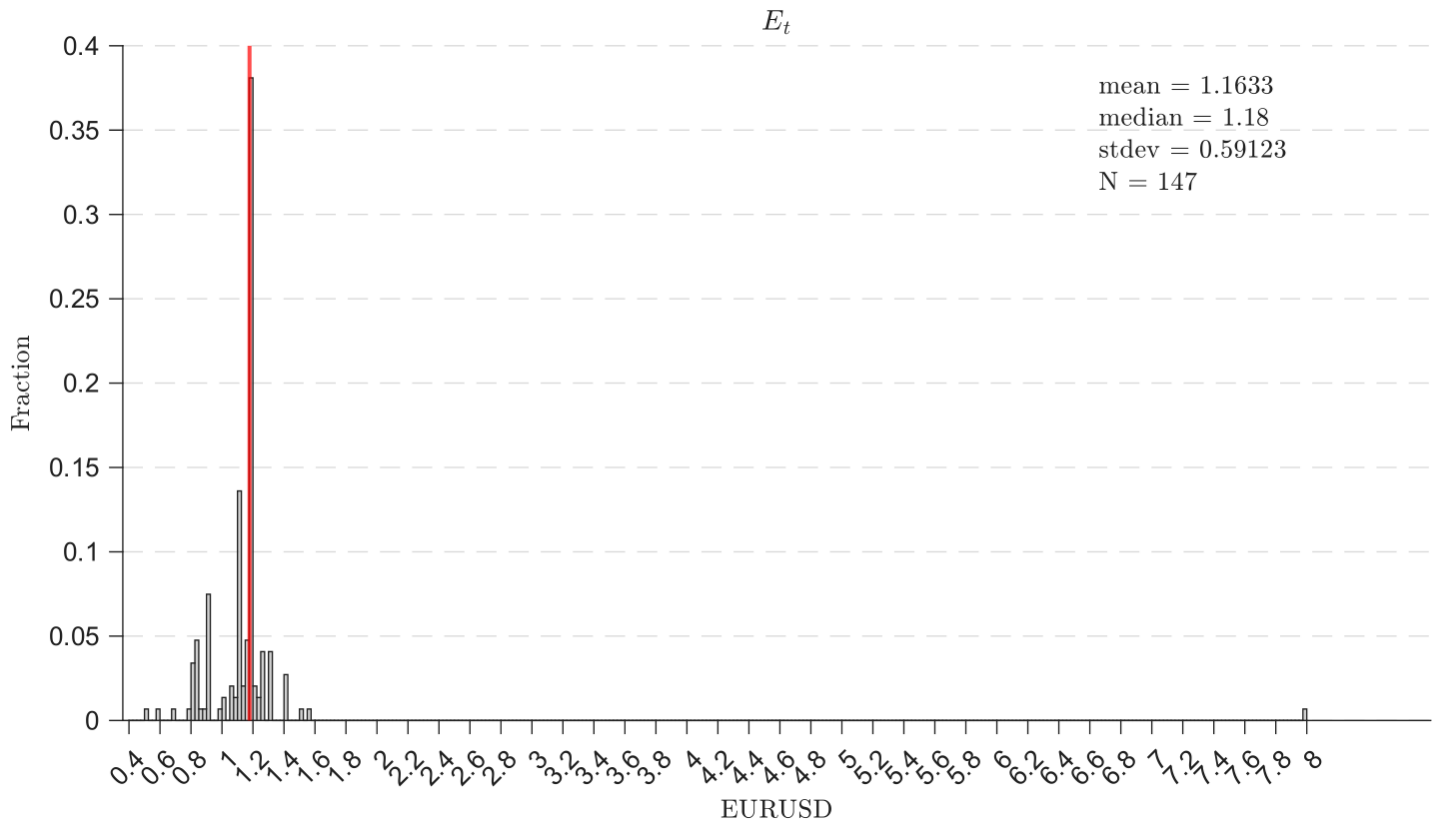
$2950 in Italy

$1999* in the US

* + 6-10% sales tax depending on the state

The Nominal Exchange Rate

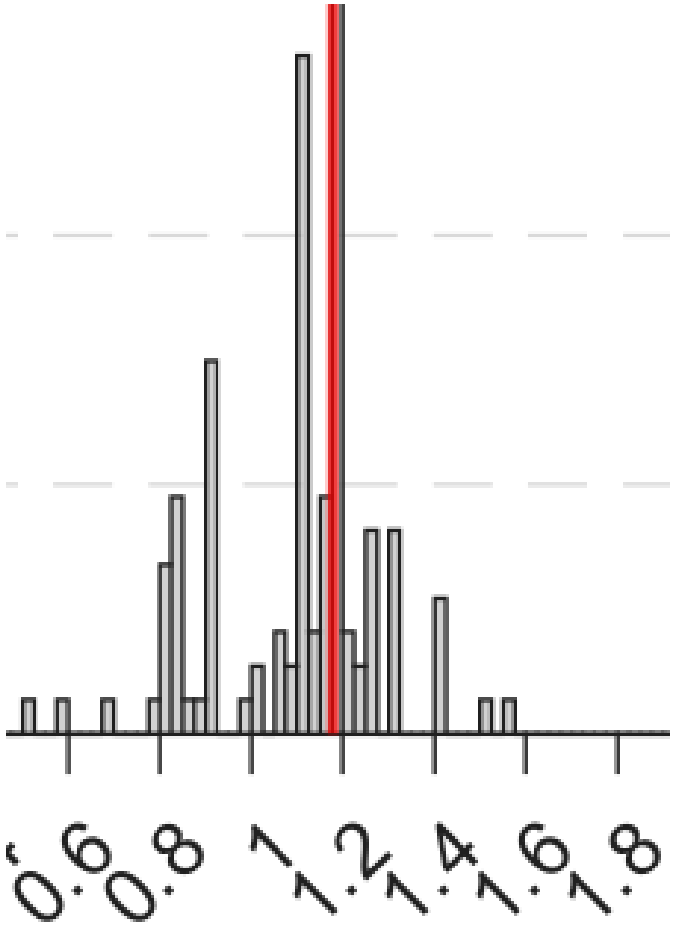
- The nominal exchange rate is the amount of currency needed to buy one unit of another currency



Q: *What is the current US Dollar-Euro exchange rate (measured in US dollars per one Euro)?*

The actual value on the quiz dates (Feb 5 and Feb 9, 2026) was 1.18.

The Nominal Exchange Rate



- Note the two spikes in your answers around 1.2 and 0.8
- They most likely reflect that some of you confused the units of account

The Nominal Exchange Rate

- The nominal exchange rate is the amount of currency needed to buy one unit of another currency
- Notation
  1. **Unit after the numbers**
     - **1.18 USD/EUR** means that you can exchange €1 for $1.18
     - **0.82 EUR/USD** means that you can exchange $1 for €0.82
  2. **Units before the numbers**
     - **USD/EUR (or USDEUR) 0.82** means that you can exchange $1 for €0.82
     - **EUR/USD (or EURUSD) 1.18** means that you can exchange €1 for $1.18
  - In class, we will denote the exchange rate E (sometimes $E_{\$/€}$) and use the "1.18 USD/EUR" notation
- Exchange rate can be quoted in two ways
  1. Direct: the price of the foreign currency in terms of the home currency
  2. Indirect: the price of the home currency in terms of the foreign currency
- In this class, we will use the direct quotation
  - How to remember? This is how we express goods prices as well
  - Some countries (UK, Canada, etc.) use the indirect way

The Nominal Exchange Rate

Cross rates

Majors

Currencies	United Kingdom GBP	United States USD	Eurozone EUR	Australia AUD	Canada CAD	China CNY	Japan JPY	Switzerland CHF
1 GBP £	1.0000	1.3631	1.1496	1.9272	1.8575	9.4163	209.18	1.0484
1 EUR €	0.8695	1.1855	1.0000	1.6761	1.6155	8.1894	181.93	0.9118
1 USD $	0.7334	1.0000	0.8434	1.4138	1.3627	6.9080	153.46	0.7691
1 AUD A$	0.5186	0.7071	0.5964	1.0000	0.9636	4.8846	108.51	0.5438

Data delayed at least 10 minutes, as of Feb 16 2026 16:10 GMT.

Source: FT on Feb 16, 2026

The Nominal Exchange Rate

- *Depreciation* of home currency
  - An *increase* in the price of foreign currency in terms of home currency (E goes up)
  - Home goods become *cheaper* for foreigners, and foreign goods become *more expensive* for domestic residents
- *Appreciation* of home currency
  - A *decrease* in the price of foreign currency in terms of home currency (i.e., E decreases)
  - Home goods become *more expensive* for foreigners, and foreign goods become *cheaper* for domestic residents

Five Facts about FX Markets

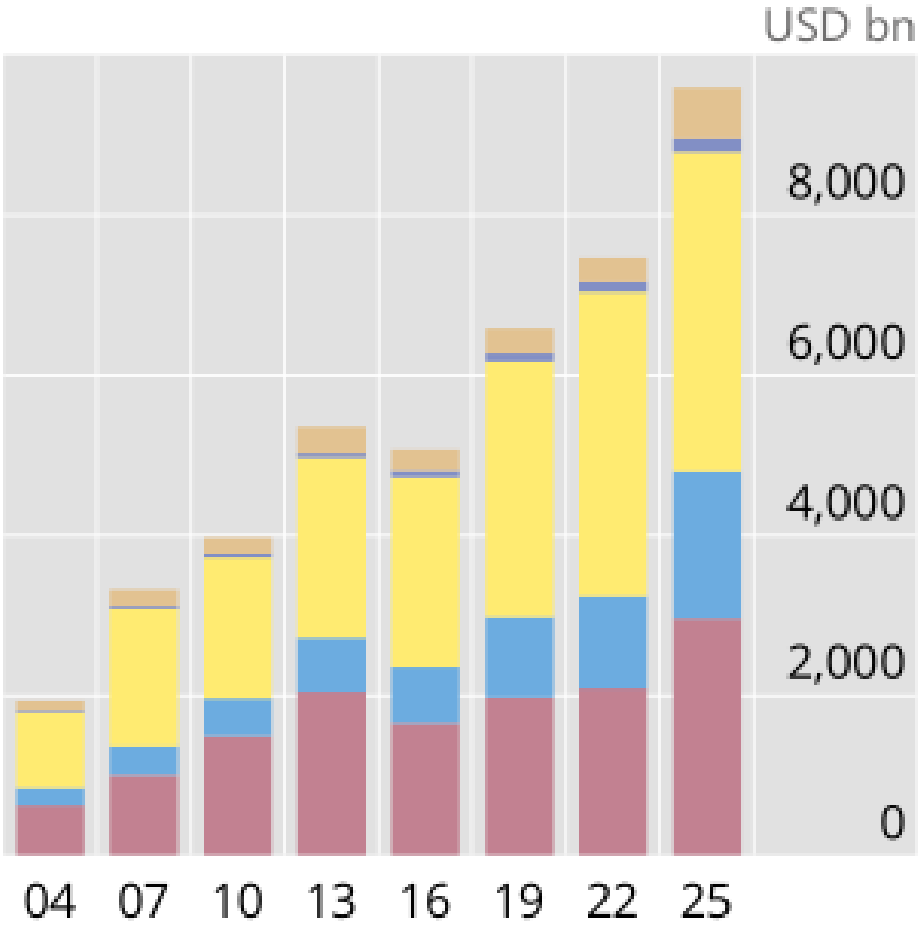
1. FX Market is Enormous

The daily volume of foreign exchange transactions was $9.6 trillion per day in April 2025

- More than 4 times Italian GDP

Global foreign exchange market turnover

Net-net basis
Daily averages, in millions of US dollars

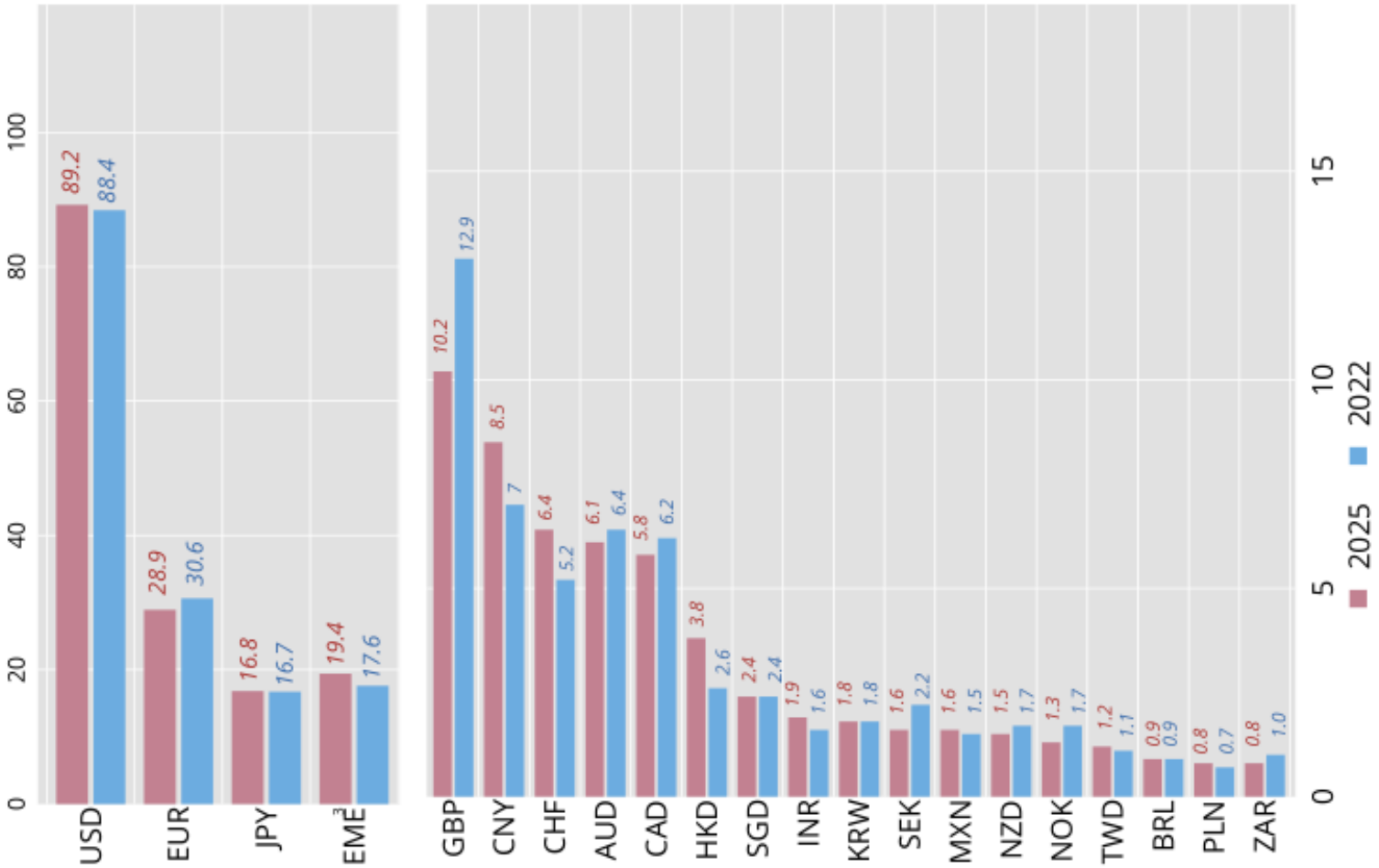


Source: BIS
Note: ignore colors. What matters, for now, is the size of each bar

2. $ is really important

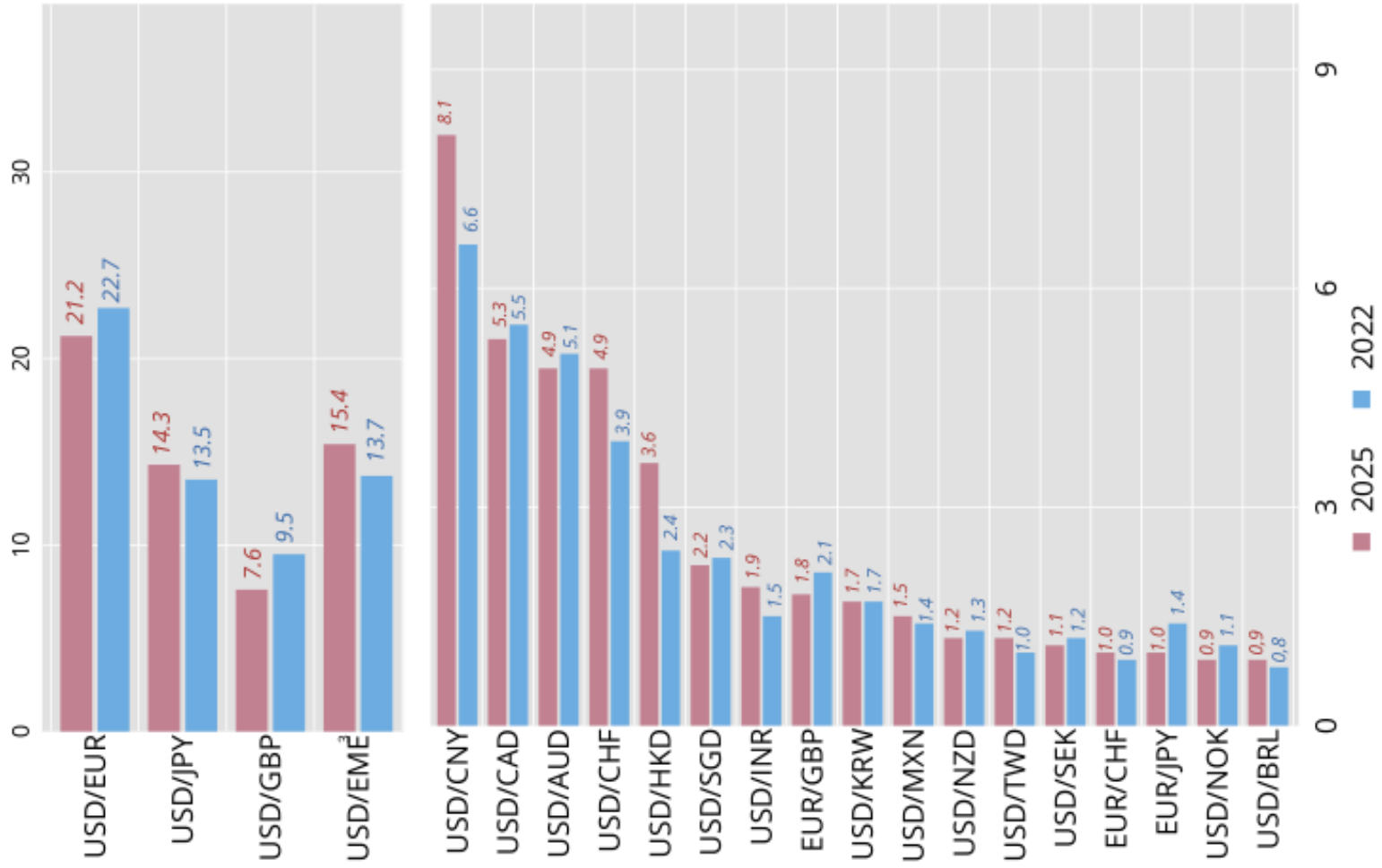
Source: BIS

Currency Distribution (percentage of daily turnover)



- Most transactions exchange foreign currencies for USD
- The prices of some of the most traded goods (such as oil) are typically stated and paid in USD
- Many trades for "smaller" currencies are made using the **vehicle currency** ($). Does it make any sense to do this?

USD is a Vehicle Currency



- USD is like money in currency transactions

3. FX is traded in Financial Centers

Top 10 FX Trading Centers — Share of Global Turnover (%)

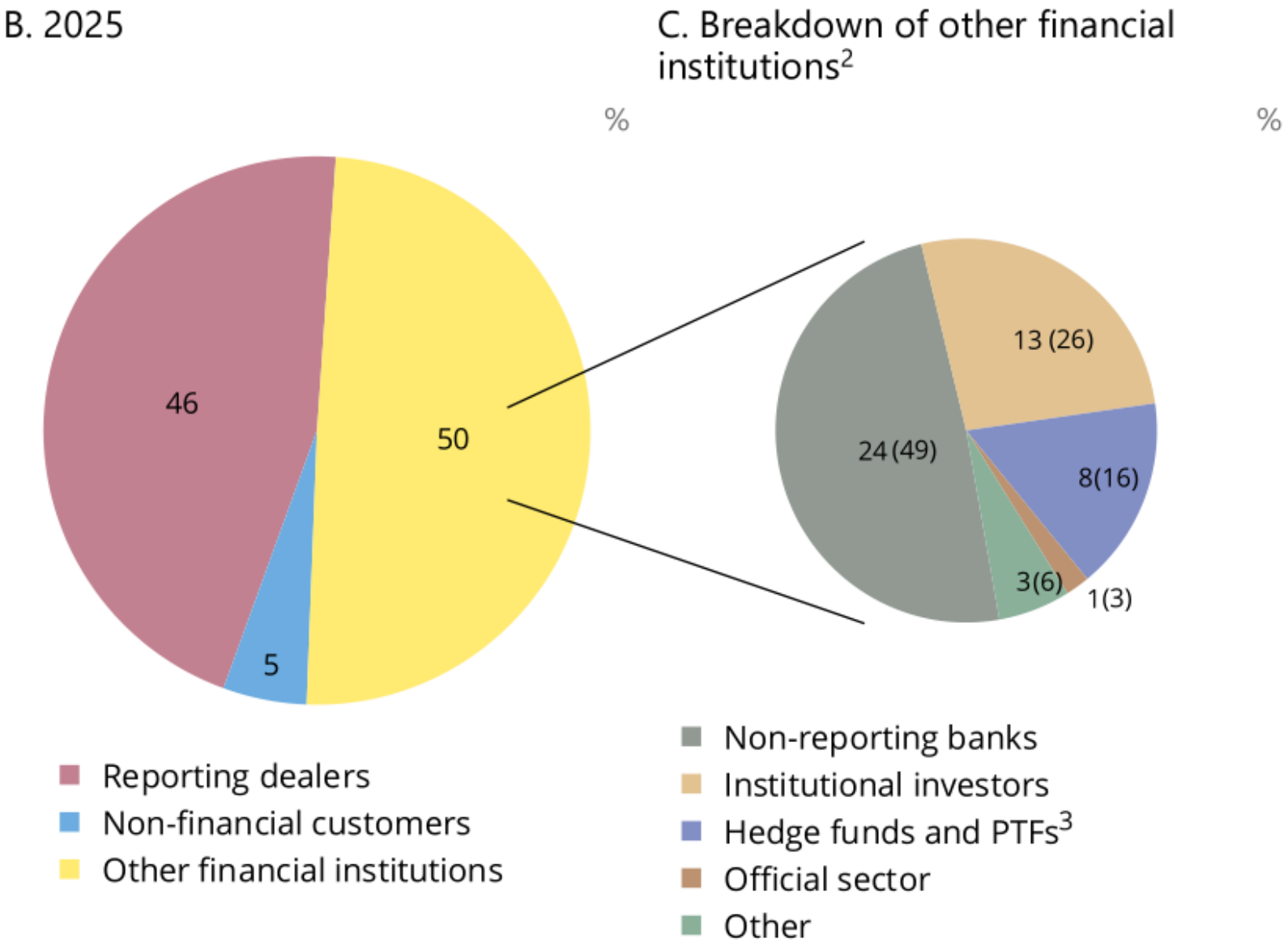
(Sorted by 2025 share)

Country	2013	2016	2019	2022	2025
United Kingdom	40.8	36.9	43.2	38.1	37.6
United States	18.9	19.5	16.5	19.5	18.6
Singapore	5.7	7.9	7.7	9.5	11.8
Hong Kong SAR	4.1	6.7	7.6	7.1	7.0
Japan	5.6	6.1	4.5	4.4	3.5
Germany	1.7	1.8	1.5	1.9	3.1
Switzerland	3.2	2.4	3.2	3.6	2.9
France	2.8	2.8	2.0	2.2	1.9
China	0.7	1.1	1.6	1.6	1.9
Australia	2.7	1.9	1.4	1.5	1.6

- The trade is concentrated in big financial centers
  - The market is always open
- Given the huge activity on FX markets, prices are quickly equalized in all major trading places in the world
  - Arbitrage: buy low and sell high

Source: BIS, Table 6 + AI

4. Financial Firms are the Key Participants



Taxonomy (no need to learn by heart)

- **Reporting dealers** – Large banks and securities firms that actively trade FX/derivatives (inter-dealer and with customers) and report to the BIS survey.
- **Other financial institutions** – Non-dealer financial firms active in FX markets, including smaller banks, mutual/pension funds, hedge funds, insurance companies, leasing companies, and central banks.
- **Non-reporting banks** – Smaller or regional banks not classified as reporting dealers.
- **Institutional investors** – Asset managers such as mutual funds, pension funds, insurers, and endowments that use FX mainly for investment and hedging.
- **Hedge funds & proprietary trading firms** – Leveraged, often speculative investors, including high-frequency trading firms.
- **Official sector financial institutions** – Public sector entities such as central banks and international financial institutions (e.g., BIS, IMF, development banks).
- **Non-financial customers** – Corporations, governments, and individuals using FX for trade, investment, or retail purposes.
- **Other** – Residual category not fitting the above groups.

Source: BIS

4. Financial Firms are the Key Participants

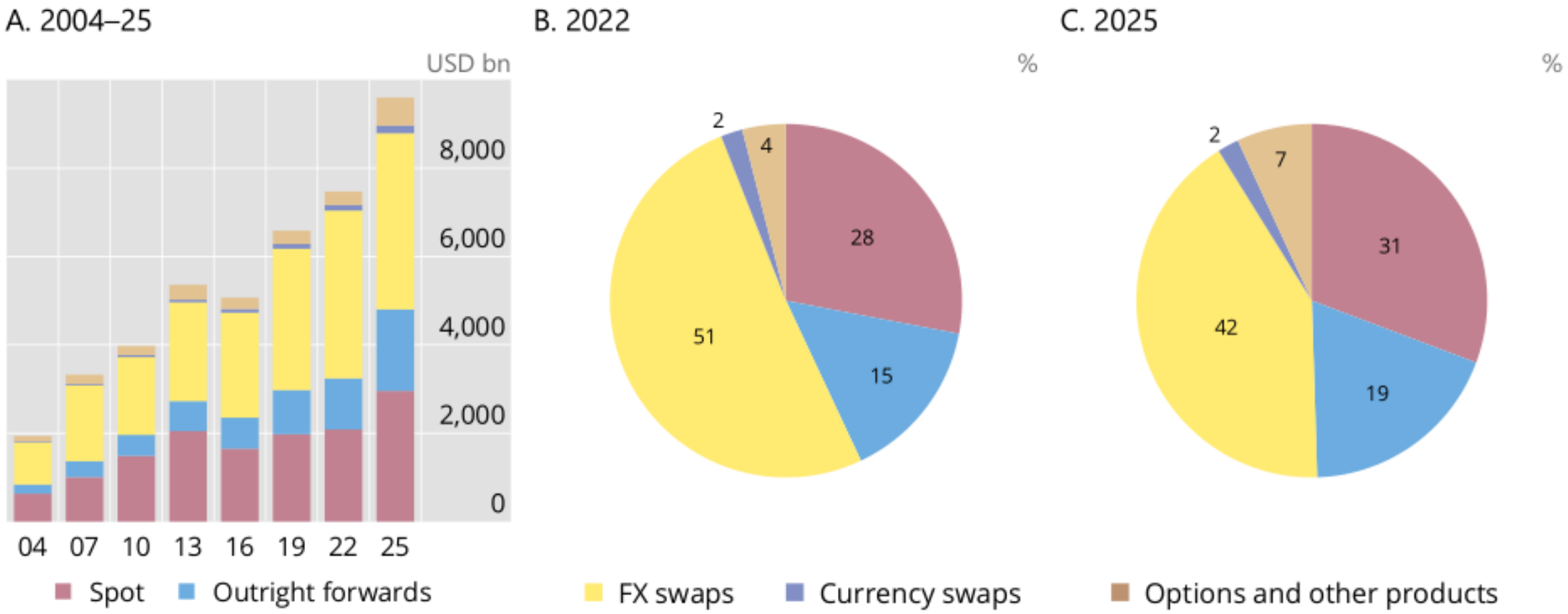
1. Financial firms (hedge funds, insurance companies, etc.)
   - Either direct speculation on currencies or need to get foreign currency to let their clients invest in foreign stocks and bonds
2. Banks (retail FX transactions)
   - Italian tourists buy dollars for their holidays in the US
   - Firms that need foreign currency to pay their foreign suppliers
3. Non-financial firms
   - Conduct foreign currency transactions to buy/sell goods, services and assets.
4. Central banks
   - conduct official international reserves transactions.

5. Spot, Forward, Swap contracts

- Spot
  - Exchange market for immediate payment and delivery (‘on the spot’)
- Forward
  - Exchange market for payment and delivery at some future date, pre-agreed upon
- Swap
  - combined spot/forward or forward/forward transactions

- Why does anybody want to use forward contracts? Swaps?

5. Spot, Forward, Swap contracts

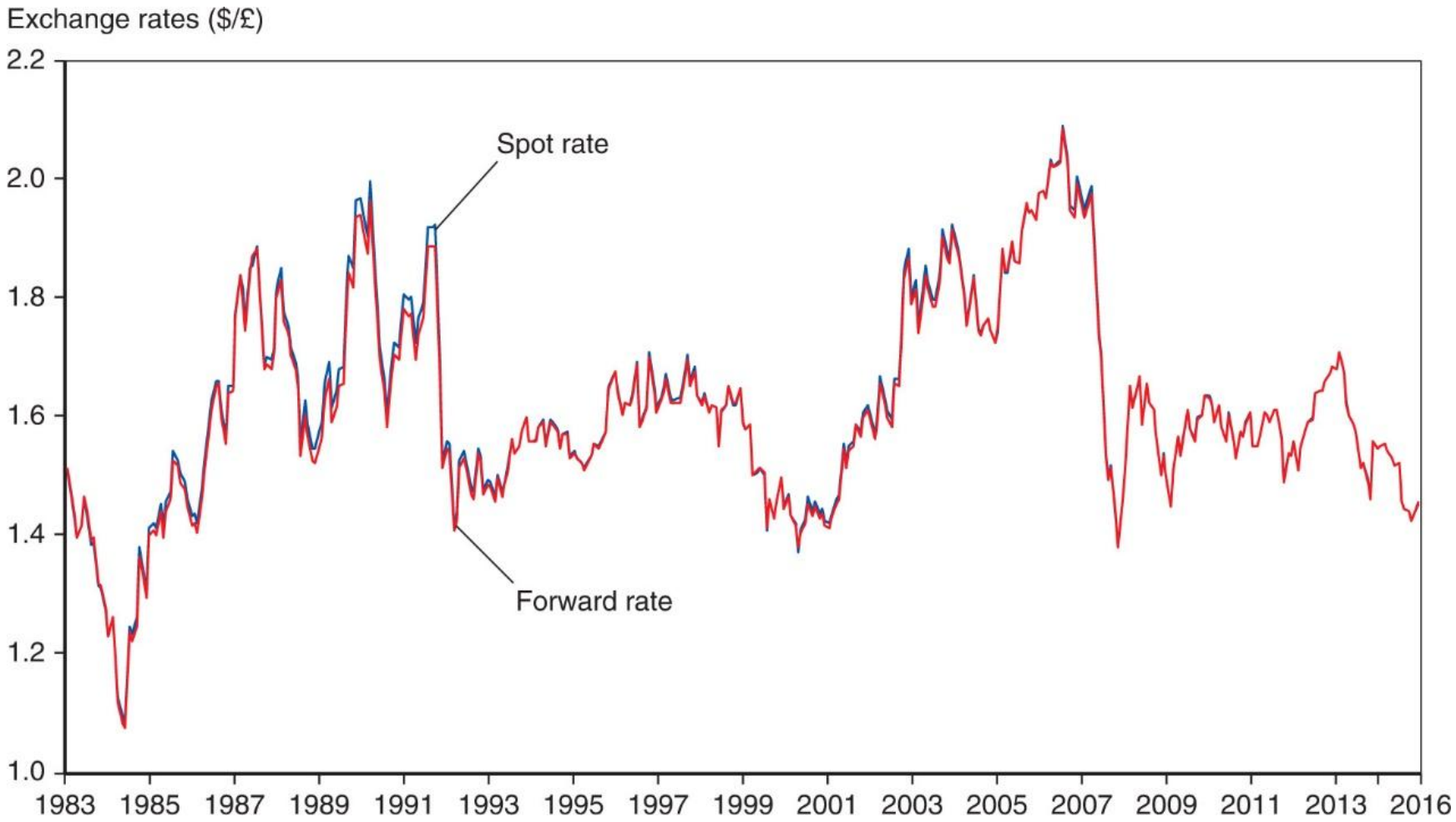


[1] Adjusted for local and cross-border inter-dealer double-counting, ie "net-net" basis.

Source: BIS Triennial Central Bank Survey. For additional data by instrument, see Table 1.

- More than half of the FX trades are (FX and Currency) swaps

5. Spot, Forward, Swap contracts



Source: Datastream. Rates shown are 90-day forward exchange rates and spot exchange rates, at end of month.

- Spot and forward exchange rates tend to move in a highly correlated fashion

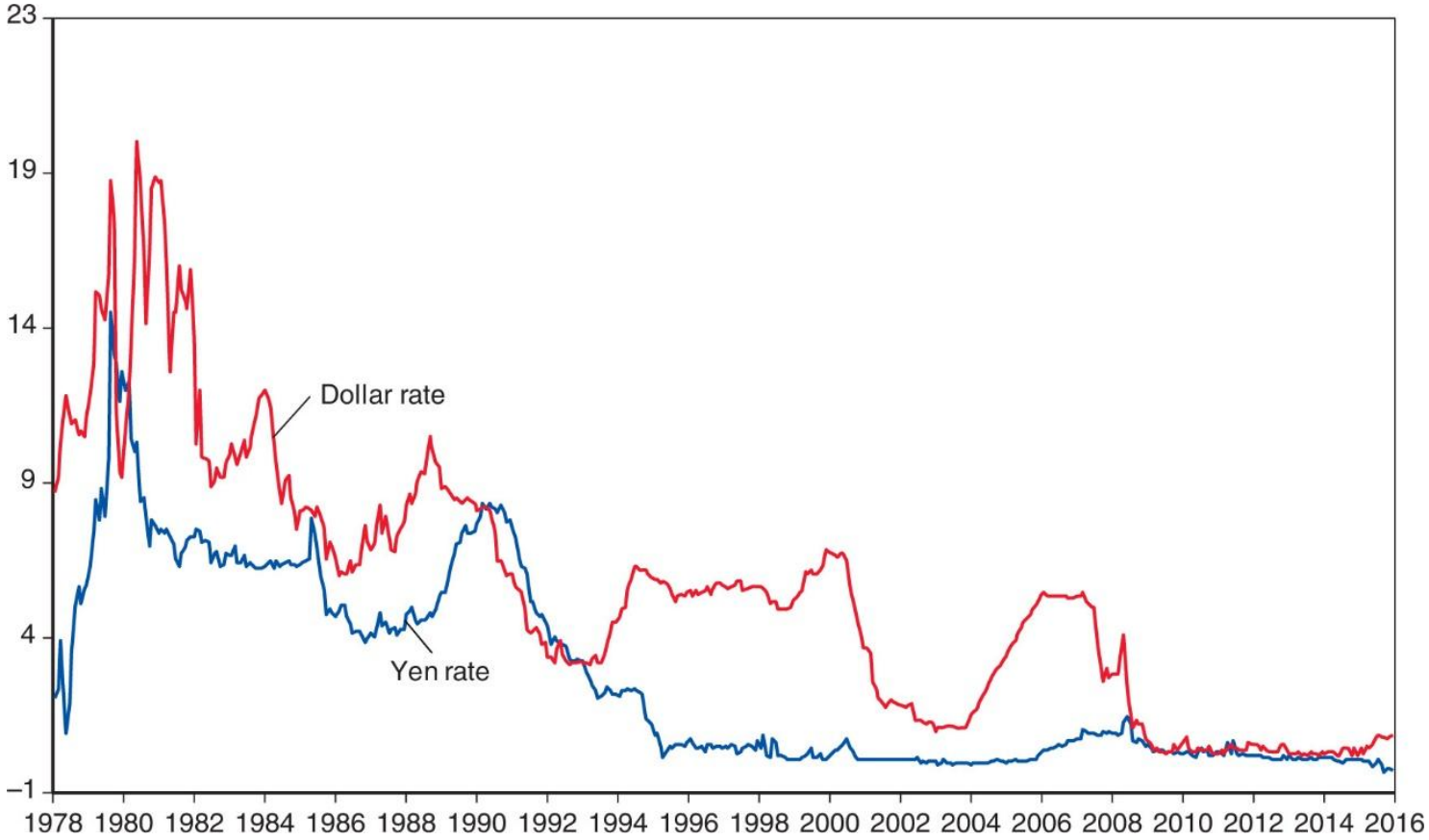
The Demand for Currency Deposits

Demand for Foreign Currency Assets

- The demand for a foreign currency bank deposit is influenced by the same considerations that influence the demand for any other asset
  - Rate of return
  - Risk
  - Liquidity

Demand for Foreign Currency Assets

Interest rates in US and Japan



Source: Datastream. Three-month interest rates are shown.

Demand for Foreign Currency Assets

- *For now*, we will assume that investors are primarily concerned about the rates of return on currency deposits

- Rates of return depend on
  - interest rates that the assets will earn
  - exchange rate growth expectations

Demand for Foreign Currency Assets

Example in which the US is a home country

- Dollar deposit interest rate is $R_{\$} = 2\%$
- Euro deposit interest rate is $R_{€} = 6\%$
- Exchange rate today is $E_{\$/€} = 1\$/€$
- Expected exchange rate in one year is $E^{e}_{\$/€} = 0.97\$/€$
- Investing in euro deposits
  - $\$100$ can be exchanged today for $€100$
  - These $€100$ will yield $€106$ after one year
  - These $€106$ are expected to be worth $€106 \times 0.97 = \$102.82 \approx \$103$

Demand for Foreign Currency Assets

An example where the US is home country

2%

$$\frac{\$102.82 - \$100}{\$100} = 2.82\%$$

Demand for Foreign Currency Assets

- Note that the expected rate of appreciation of the euro or, equivalently, the rate of depreciation of the dollar is

$$\frac{E^e - E}{E} = \frac{0.97 - 1}{1} = -3\%$$

- The dollar rate of return on Euro deposits ***approximately*** equals:
  - the interest rate on euro deposits
  - **plus** the expected rate of appreciation of € or depreciation of $

$$6\% + (-3\%) = 3\%$$

Expected rate of return on a foreign deposit $\approx R_{€} + \frac{E^e - E}{E}$

Equilibrium on the Forex Market

- Equilibrium
  - deposits of all currencies offer the same expected return

- We will call it **the interest (rate) parity**

Equilibrium on the Forex Market

$$R = R^* + \frac{E^e - E}{E}$$

Why should this condition hold?

- Assume $R < R^* + (E^e - E)/E$
- No investor wants to hold home deposits
- Demand for deposits in domestic currency ↓
- Demand for deposits in foreign currency ↑
- Home currency depreciates $\boldsymbol{E}$ ↑ (assuming that E^e is fixed)

Back to the Numerical Example

$$R = R^* + \frac{E^e - E}{E}$$

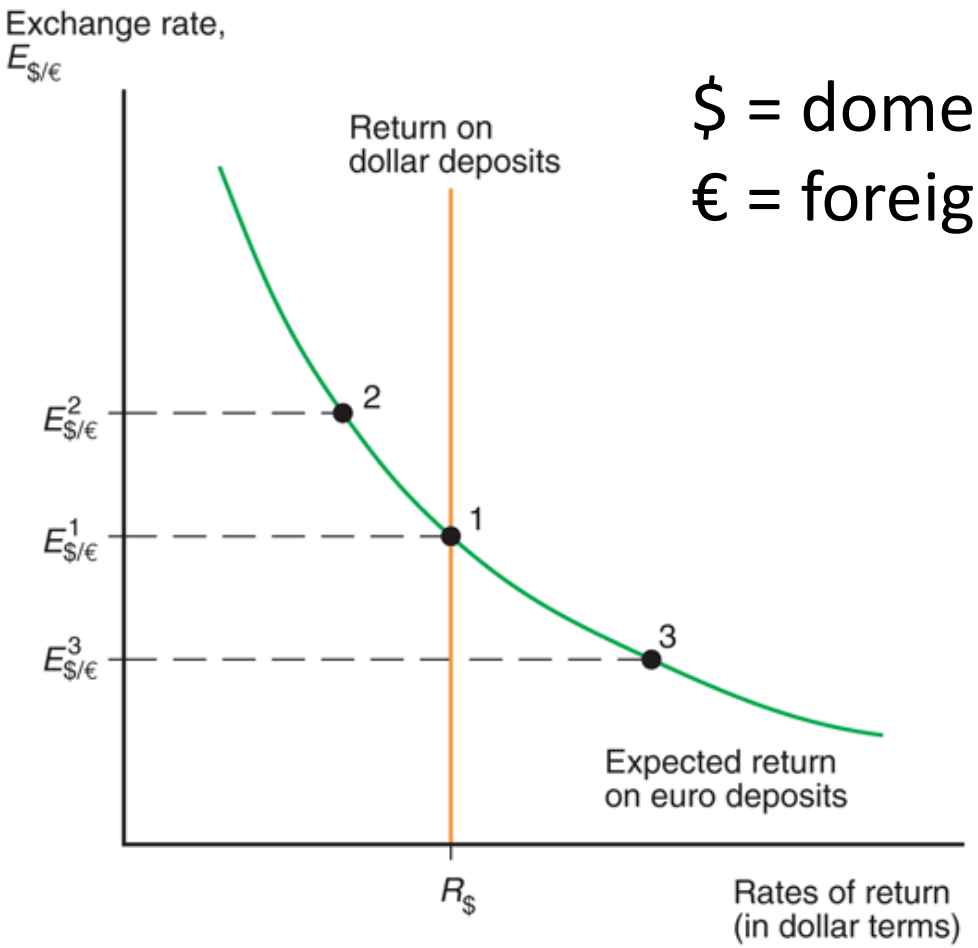
- In our example: $E = 1; E^e = 0.97; R_{\$} = R = 2\%; R_{€} = R^* = 6\%$

- This imlies $R < R^* + (E^e - E)/E$ because $2\% < 3\%$

- The dollar (domestic currency) must depreciate $(\mathrm{E} \uparrow)$ to a new level $E \approx 1.01$

A Useful Diagram

A Convenient Way to Represent the Parity



$ = domestic currency

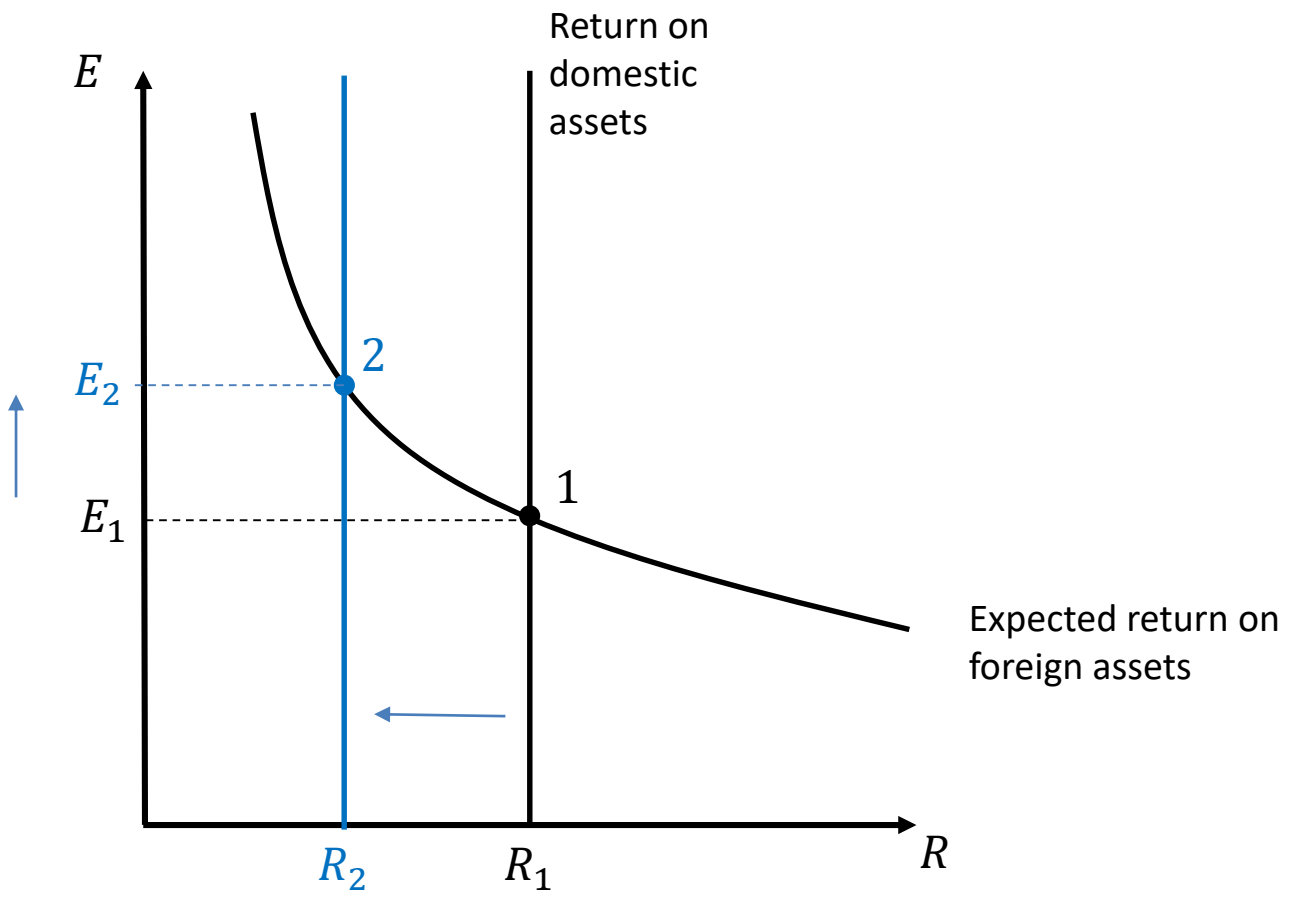
€ = foreign currency

$$R_\$ = R_€ + \frac{E^e - E}{E}$$

Note: $R_€$ and E^e are fixed on this diagram

Exercise 1

A Decrease in Domestic Interest Rate



At E_1, the expected rate of return on foreign assets is higher than R_2

$\Downarrow$

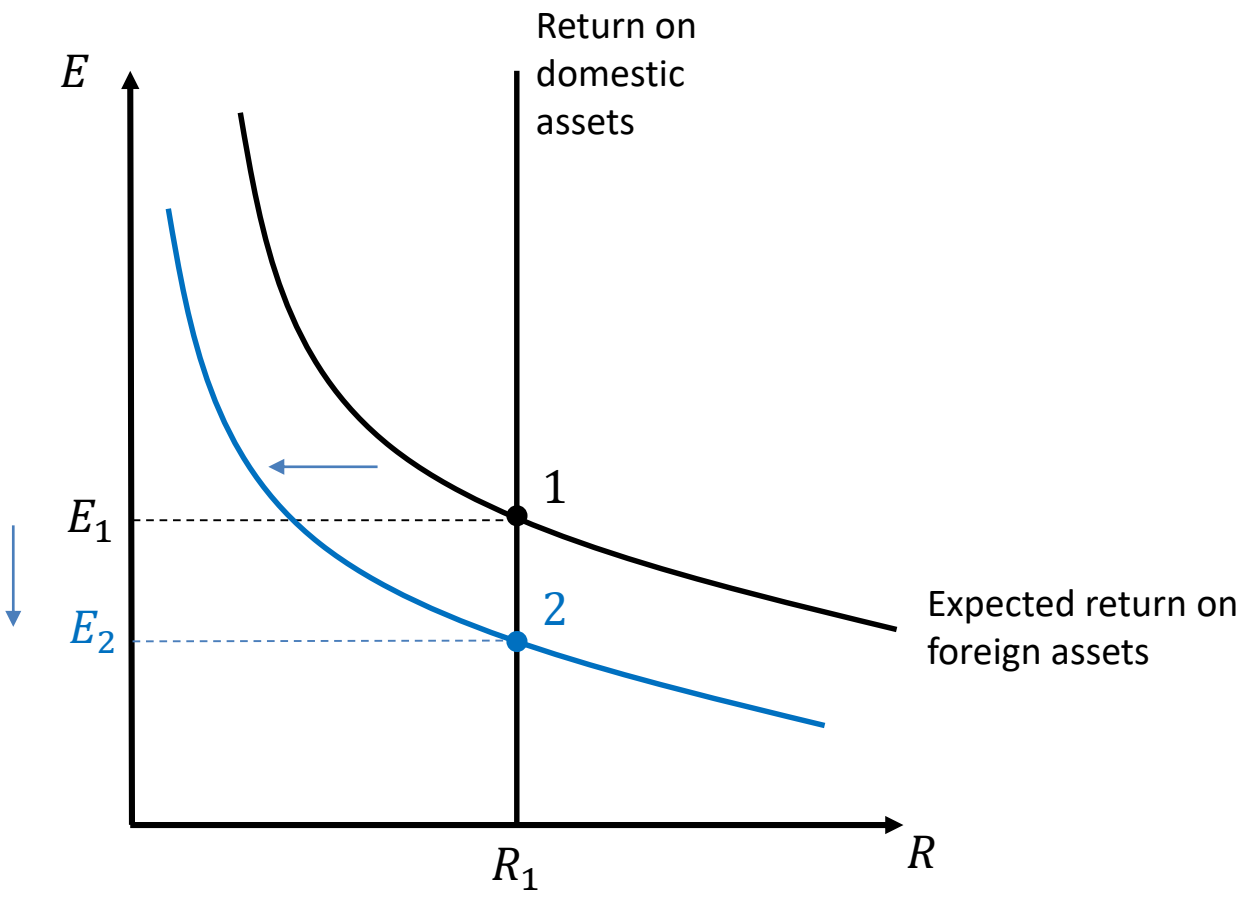
higher demand for foreign currency

$\Downarrow$

the domestic currency depreciates

Exercise 2

A decrease in the foreign interest rate



At E_1, the expected rate of return on foreign assets is lower than R_1

$\Downarrow$

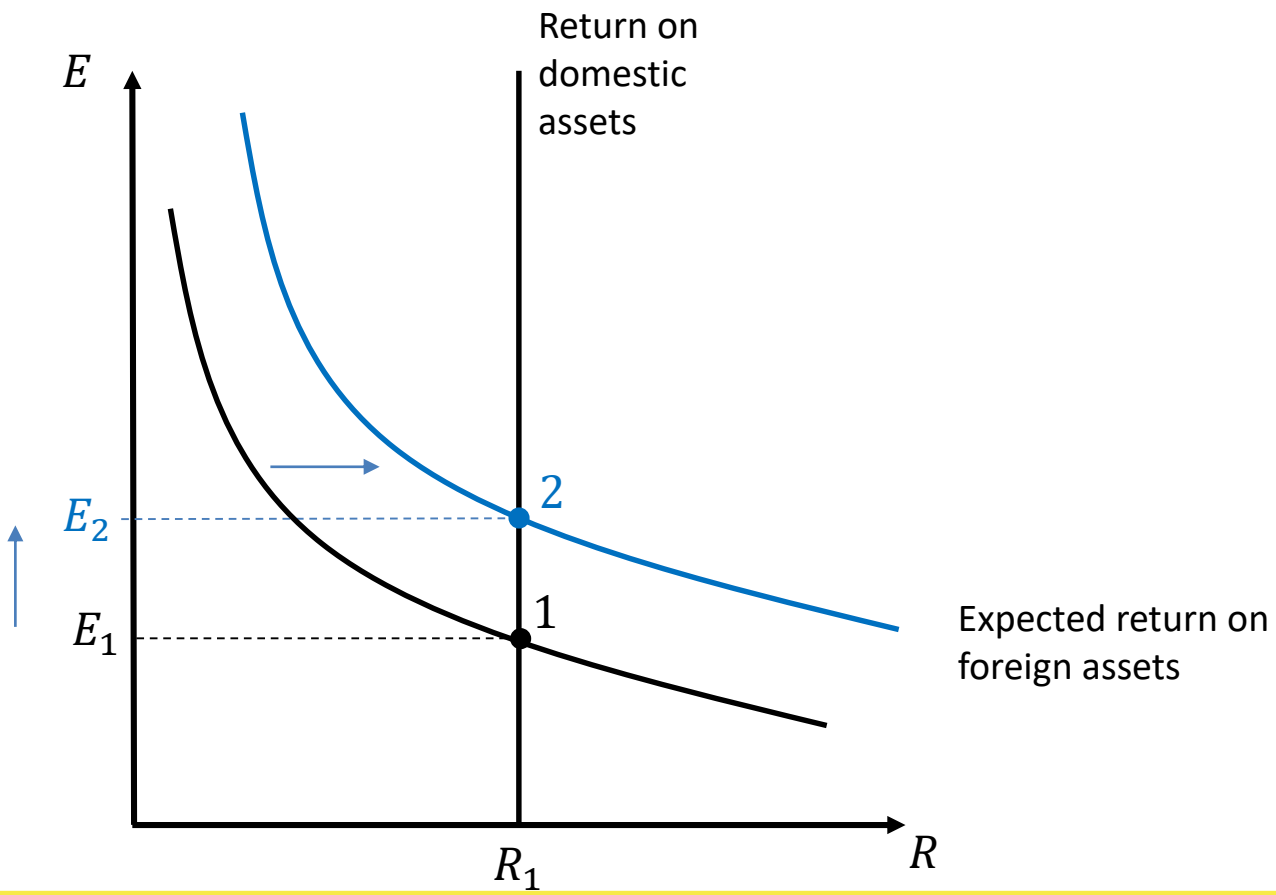
higher demand for domestic currency

$\Downarrow$

the domestic currency appreciates.

Exercise 3

An increase in E^e



$E^e \uparrow$

$\Downarrow$

higher demand for foreign currency

$\Downarrow$

the foreign currency appreciates/the domestic currency depreciates

The exchange rate behaves like any other asset price: its value today depends on expectations of its value in the future

Interest Parities

1. Covered interest parity (CIP)

$$R = R^* + \frac{F - E}{E}$$

   – F is the forward exchange rate
   – Forward exchange contract covers the FX risk

2. Uncovered interest parity (UIP)

$$R = R^* + \frac{E^e - E}{E}$$

   – Uncovered because exposes to exchange rate risk

Next Lecture

- KOM, Chapter 14 and 15